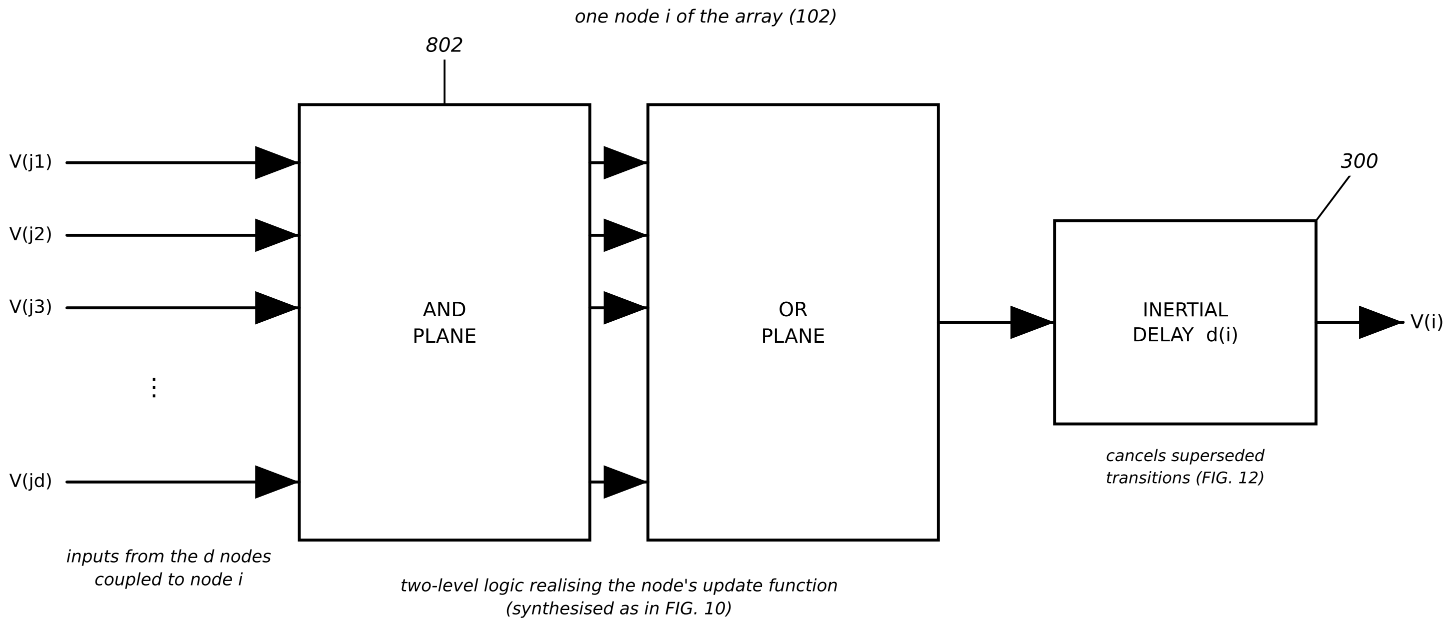
NODE CIRCUIT AND THE DELAY SIZING RULE



V(i) becomes an input to every node coupled to i, and to no other node. Node i does not receive its own output.

SIZING RULE FOR THE NODE DELAY

$$d(i) > t_{pd,max}(\text{ node } i \text{ }) \quad \text{and} \quad d(i) \neq d(j) \text{ for every coupled } j$$

The first inequality is what makes the second one mean anything. If the node delay is shorter than the worst-case propagation delay through that node's own logic, commits interleave in an order the delay values no longer determine and the class ordering is inoperative — the array still runs, and still settles, but not under the schedule that was designed.

The two inequalities are independent. Satisfying only the second gives a circuit that violates its own schedule; satisfying only the first gives a correctly-timed circuit with no ordering at all.

The delay element is the same element as FIG. 3. When its rejection window is programmable, one element performs both functions: it sequences the node and it sets how much hazard activity is admitted.

FIG. 8